Turning on the real voiceover

The build picks its speech engine automatically, in this order:

ORDER	ENGINE	TRIGGER	OUTPUT
1	**Gemini TTS** (AI Studio)	`GEMINI_API_KEY`	Publishable, muxed into the ads
2	Google Cloud TTS	`GOOGLE_TTS_KEY`	Publishable, muxed into the ads
3	espeak + MBROLA	nothing set	Guide track, stamped, separate file

Prefer Gemini. It takes a natural-language direction alongside the line, and the two concepts want opposite reads — a wry founder recounting what he found, against a flat voice reading out a measurement. Those directions live in `vo-style.json` and are the main quality lever here; the voice matters less than the direction given to it.

Why Google and not the others

Probed from this environment, 22 Sep 2026:

SERVICE	REACHABLE
Gemini / AI Studio API (`generativelanguage.googleapis.com`)	**yes**
Google Cloud TTS (`texttospeech.googleapis.com`)	**yes**
AI Studio web UI (`aistudio.google.com`)	no — and it is a UI, not an API
MiniMax (`api.minimax.io` , `api.minimaxi.chat`)	no
ElevenLabs	no
OpenAI	no
Cartesia · Deepgram · PlayHT	no

The blocked ones fail at the network layer, before any credential is checked, so a key for them changes nothing on its own. To use one, add its API host to the allowed hosts in this Claude Code environment's settings first.

Setup — Gemini (recommended)

An AI Studio **page URL is not a credential**. The key is a separate thing.

1. Go to https://aistudio.google.com/apikey and create an API key on the project you want billed. Free tier covers this volume comfortably.
2. Add it to this environment's variables as `GEMINI_API_KEY`. Environment variables are read at session start, so a session already running will not see it — start a new one.
3. Confirm, then build:

`python3 vo.py --models # what this key actually serves python3 vo.py # should print "engine: gemini" NODE_PATH=/opt/node22/lib/node_modules node build.js`

`--models` matters because preview model ids move. The default is `gemini-3.1-flash-tts-preview`; if the key does not serve it, the error says so and `GEMINI_TTS_MODEL` overrides it.

Setup — Cloud TTS (alternative)

1. Enable **Cloud Text-to-Speech API** on the project and create an API key.
2. Set it as `GOOGLE_TTS_KEY`.
3. `python3 vo.py --voices` lists en-GB, en-US and ar-* voices, then build as above.

The four motion posts rebuild with narration inside them. The `*-guide-vo.mp4` files are deleted automatically, because a guide is only needed while the read is synthetic.

The direction controls duration, not just tone

Measured on the same line, same voice, same model:

DIRECTION	"EVERY ONE OF THEM IS RUNNING ADS RIGHT NOW."
"...unhurried..."	6.12s
"...normal conversational speed, no dramatic pauses, this is a 25-second social video..."	3.60s

Same words, **41% shorter**. Across the whole script that was the difference between a 49-second TikTok ad and a 32-second one — and under 30 seconds is where short-form cold traffic actually performs, so the first version was unusable regardless of how good it sounded.

The lesson for anyone editing `vo-style.json`: **state the format and the pace, not only the mood.** "Unhurried" is a reasonable instruction for tone and a bad one for a feed. Every style string here names the target length on purpose. After editing, check the totals `vo.py` prints before rebuilding.

Rate limits, measured

On the free tier, `gemini-3.1-flash-tts-preview` allows **3 requests per minute**. The script narrates line by line, so a full run is 16 requests and takes about six minutes. Three things make that painless:

- **Throttling.** Calls are spaced to stay inside the limit rather than hammering it and backing off. `GEMINI_TTS_RPM` raises it on a paid key.
- **Retry.** A 429 is caught, the delay Google names in the response is honoured, and the line retries — up to six times.
- **A cache.** Each line is keyed on its text, style, voice, model and engine. A re-run only pays for lines that actually changed, so fixing one word in the script costs one request, not sixteen. `vo/.cache.json` holds it; delete it to force a full regenerate.

Measured on this key, 22 Sep 2026:

MODEL / API	FREE-TIER VERDICT
`gemini-2.5-flash-preview-tts`	**Works.** Quota covers a full 16-line run. The default.
`gemini-3.1-flash-tts-preview`	Caps around **10 requests a day**. One script is 16, so it cannot finish a run on the free tier. Marginally nicer read; worth switching to on a paid key.
`gemini-2.5-pro-preview-tts`	Quota exhausted immediately.
Cloud TTS (`texttospeech.googleapis.com`)	**Rejects API keys entirely** — HTTP 401, "API keys are not supported by this API. Expected OAuth2 access token". Needs a service account, not an AI Studio key.

So on a free AI Studio key there is exactly one workable route, and it is the default. Enabling billing on the project removes the cap and opens 3.1.

A two-word probe is not a valid test. `gemini-2.5-flash-preview-tts` returned no audio for the text "Quota probe." and full audio for a real sentence. When checking whether a model works, send a real line.

Security note

The key is sent as an `x-goog-api-key` header, never as a URL query parameter, so it cannot leak through a log line, a proxy access log or a traceback. Keep it in the environment. `.gitignore` covers `.gemini_key`, `*.key` and `.env`; nothing in this repository contains a credential.

Knobs

VARIABLE	DEFAULT	NOTES
`GEMINI_API_KEY`	—	Preferred. `GOOGLE_AI_STUDIO_KEY` also works.
`GEMINI_TTS_MODEL`	`gemini-3.1-flash-tts-preview`	Verify with `--models`; preview ids move.
`GEMINI_TTS_VOICE`	`Charon`	A prebuilt Gemini voice. If the name is wrong the API says so.
`GOOGLE_TTS_KEY`	—	Cloud TTS instead. `GOOGLE_API_KEY` also works.
`GOOGLE_TTS_VOICE`	`en-GB-Neural2-D`	Run `--voices` and pick from what the account actually serves rather than trusting this default.
`GOOGLE_TTS_RATE`	`0.96`	Slightly under natural pace; the ad's argument is measurement, so an unhurried read suits it.

Scene lengths are re-measured from whichever engine ran, so the picture re-cuts itself to the new read. Nothing needs timing by hand.

Two things a better voice does not fix

Concept A speaks in the first person — "I filled in the contact form on N Dubai businesses". The campaign's whole argument is that the claim is literally true and independently checkable. A synthetic voice reading it is a

small dishonesty at the exact point the campaign asks to be trusted, and a good synthetic voice makes that worse rather than better, because nobody can tell. Recommend the founder records it: sixty seconds of phone audio. Concept B makes no first-person claim, so synthetic narration is fine there.

Arabic still needs a human. Google serves `ar-XA` voices and will read the Snapchat copy cleanly, but reading a draft well is not the same as the draft being right. That copy still needs a native Gulf-dialect writer before it runs.